

SHA-1 Hash Function Demonstration

SHA-1 takes any input and produces a fixed 160-bit (40 hex characters) hash value. Try different inputs to see how even small changes produce completely different hashes.

Plaintext (string):

test

Calculate SHA-1

SHA-1 Hash output (hex):

a94a8fe5ccb19ba61c4c0873d391e987982fbbd3

HMAC Construction (Simplified Educational Version)

HMAC combines a hash function with a secret key to provide both data integrity and authentication. This simulation uses a simplified "dummy" hash function for educational purposes.

Step 1: Setup Parameters

Message (binary):

Binary message to be authenticated

10100111010100011011001100100111000100110010001010100011010010110001
1010110101010001100001010

Generate Random Message

Block size (l):

Hash function block size (must be ≥ 8)

8

Initialization Vector (IV):

Random initialization vector of length l

11111011

Generate Random IV

Secret Key (k):

Secret key of length l

00111110

Generate Random Key

Step 2: Test the Dummy Hash Function

Our simplified hash function XORs adjacent bit pairs: $H(b1b2b3b4) = (b1 \oplus b2)(b3 \oplus b4)$

Test Input (2l bits):

Input must be exactly 2l bits (16 bits for l=8)

1100101011001010

Calculate Hash

Hash Output (l bits):

Result of applying the hash function

00110011

Step 3: HMAC Calculation Guide

HMAC follows this process:

1. $ipad = 01011100$, $opad = 00110110$
2. Inner hash: $H(IV \parallel (k \oplus ipad) \parallel padded_message \parallel length)$
3. Outer hash: $H(IV \parallel (k \oplus opad) \parallel inner_hash_result)$

 Calculation Helper:

HMAC Calculation Steps:

1. Key: 00111110
2. ipad: 01011100
3. opad: 00110110
4. $k \oplus ipad$: 01100010
5. $k \oplus opad$: 00001000
6. Padded message:
101001110101000110110011001001110001001100100010101000110100101100011010110101010001100001010000
7. Length in binary: 01100000
8. Inner hash start: $H(11111011 \parallel 01100010) = 00101101$

Continue the calculation manually following the HMAC algorithm...

Show
Step-by-
Step
Calculation

 Step 4: Enter Final HMAC Result

Final HMAC Tag:
Enter the final HMAC authentication tag

01010000

Check Answer

 Summary of Results

Parameter	Value
Message	101001110101000110110011001001110001001100100010101000110100101100011010110101010001100001010000
Block Size (l)	8
Secret Key	00111110
Initialization Vector	11111011
Final HMAC Tag	01010000

Update Summary